

DETECTION OF ABANDONED OBJECTS IN VIDEO SURVEILLANCE SYSTEMS: A COMPARATIVE ANALYSIS OF RULE-BASED AND AI-ORIENTED APPROACHES

Vitalii-Oleksandr Pastukh, Anastasiia Deineko

IT STEP University, 83A, Zamarstinivska str., Lviv, 79019, Ukraine

Authors' e-mails: pastuh.vitalik98@gmail.com, anastasiya.deineko@gmail.com

<https://doi.org/10.23939/acps2026.01.091>

Received: 12.03.2026

Accepted: 27.04.2026

Published: 29.05.2026

© Pashukh V-O., Deineko A., 2026

Abstract: This review paper provides a comprehensive comparative analysis of abandoned object detection algorithms, specifically contrasting rule-based approaches with artificial intelligence (AI-based) models. The manuscript synthesizes existing research, technical documentation, and publicly available benchmark data to evaluate the applicability of these approaches in dynamic video surveillance environments. The study does not claim original experimental testing of YOLO26 or RT-DETRv2 models; instead, it analyzes reported benchmark results, architectural characteristics, and practical deployment constraints. As a review paper, this manuscript provides an analysis and classification of contemporary automated object detection methods in video streams, categorized by their operational principles and application constraints. Specifically, the study systematizes the key differences between background subtraction-based detection and modern neural network architectures, enabling the determination of optimal application domains based on available computational resources and specific environmental scenarios.

Index terms: video surveillance, abandoned objects, comparative analysis, rule-based methods, artificial intelligence, deep learning, AOD, YOLO, RT-DETR, transformers, computational complexity, public safety.

I. INTRODUCTION

The relevance of this problem is driven by the conditions of rapid urbanization and the escalation of global security threats; thus, the automation of public space monitoring is becoming a vital necessity rather than merely a technological advantage. Video surveillance systems cover thousands of square meters in transportation hubs, shopping centers, and government institutions, generating volumes of data that are physically impossible to process using human resources alone. The primary challenge in this field is Abandoned Object Detection (AOD), which is critical for preventing terrorist acts, identifying suspicious packages, and ensuring general public order. Traditionally, reliance on the human factor leads to a significant number of errors,

making automated analytical algorithms the only reliable solution for long-term monitoring.

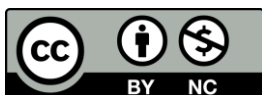
The problem of detecting abandoned objects in a video stream extends beyond the simple detection of static items. It requires a profound understanding of the scene context: it is necessary to distinguish between objects temporarily placed nearby by an owner and those intentionally left unattended. The classical definition of an abandoned object is based on spatio-temporal criteria: the item must remain stationary for a specific duration and be located at a certain distance from any individual who can be identified as the owner. Implementing these criteria in complex, dynamic environments with fluctuating lighting, dense crowds, and frequent occlusions represents an extraordinarily difficult scientific and technical task.

At the current stage of technological development, two main approaches to solving the AOD task have emerged: rule-based approaches, which are founded on mathematical background subtraction models and logical rules, and AI-oriented approaches, which leverage the power of deep neural networks for object recognition and tracking. Each of these approaches possesses its own distinct advantages and limitations.

Therefore, the present manuscript is explicitly positioned as a review paper. Its contribution lies in the systematization and comparative interpretation of existing approaches to abandoned object detection rather than in the presentation of new experimental results. The quantitative comparisons of YOLO26 and RT-DETRv2 discussed in this paper are based on publicly available benchmark data and technical documentation, while the practical conclusions are derived from their architectural properties, reported performance metrics, and suitability for different video surveillance scenarios.

II. LITERATURE REVIEW AND PROBLEM STATEMENT

Rule-based AOD systems rely on deterministic processing of pixel intensities and geometric blob



analysis. The key mechanism here is the adaptive dual background model. Two background models are maintained for each scene: Long-term model (B_L): updated with a very low learning rate ($\lambda_L \approx 0.001$). It represents the stable structure of the scene (walls, floors). Short-term model (B_S): updated quickly ($\lambda_S \approx 0.1$). It adapts rapidly to new static objects, making them part of its background representation.

Mathematically, the static foreground detection process (SF) can be represented as a logical operation on the foreground masks of both models:

$$SF(x, y, t) = F_L(x, y, t) \cap \neg F_S(x, y, t), \quad (1)$$

where M_L and M_S are binary masks obtained through threshold comparison of the current frame with B_L and B_S respectively [1].

The Gaussian Mixture Model (GMM) and its improved version, MOG2, implemented in libraries such as OpenCV, remain the foundational tools in this approach.

The standard GMM algorithm models each pixel value over time as a mixture of K Gaussian distributions. The probability of observing a pixel value X_t at time t is described by the equation:

$$P(X_t) = \sum_{i=1}^K \omega_{i,t} \cdot \eta(X_t, \mu_{i,t}, \Sigma_{i,t}), \quad (2)$$

Where $\omega_{i,t}$ is the weight of the component, and η is the Gaussian probability density function. In AOD tasks, the learning rate α is a critical parameter. If α is high, a static object (e.g., an abandoned bag) integrates into the background model too quickly and ceases to be detected as an anomaly. Conversely, an excessively low α leads to false positives during lighting changes or tree movement. [2].

However, for systems requiring high real-time performance, the ViBE (Visual Background Extractor) algorithm is often applied. Unlike GMM, which uses parametric modeling (a mixture of Gaussian distributions), ViBE is based on a non-parametric approach. For each pixel, a set of N previously obtained color samples is stored. The detection process in ViBE is described by comparing the current pixel value v with its model $M = \{v_1, v_2, \dots, v_N\}$. A pixel is classified as background if the number of samples within a sphere of radius R around it exceeds a threshold min :

$$\#\{v_i \in M : dist(v, v_i) < R\} \geq min, \quad (3)$$

The key advantage of ViBE in AOD tasks is its strategy of random updating and spatial propagation. When a pixel is identified as background, it updates not only its own model but also the model of one of its eight neighboring pixels with a probability of $1/\phi$. This allows the system to effectively fill "ghosting" artifacts that occur after objects begin moving and ensures the spatial coherence of foreground masks.

In AOD tasks, the update rate is a critical parameter for both GMM and ViBE. For GMM, this is the

parameter α , while for ViBE, it is the sample size N and the update probability $1/\phi$. If the update rate is high, a static object integrates into the background model too quickly and ceases to be detected as an anomaly. Conversely, an excessively low rate leads to false positives during lighting changes or in the presence of a dynamic background (e.g., tree leaves) [3].

AI-oriented approaches replace manual feature engineering with the automatic learning of hierarchical representations. A typical modern AOD architecture consists of three levels:

1. Detection Level (Backbone + Head): Models from the YOLOv8 family are utilized, which are based on the CSPDarknet53 architecture. They are capable of recognizing not only the presence of an object but also its semantic class (backpack, bag, suitcase, box) with high precision, even under conditions of partial occlusion [4].

2. Tracking Level: The DeepSORT algorithm integrates a Kalman filter for motion prediction and deep appearance descriptors for object identification in consecutive frames. This allows for maintaining the "owner-object" connection even if they temporarily disappear from the field of view.

3. Spatiotemporal Modeling Level: ConvLSTM layers are employed, applying convolutional operations to internal LSTM transitions. This enables the model to understand that an abandoned object is not merely a static spot, but the result of a specific dynamic action captured in previous frames [5].

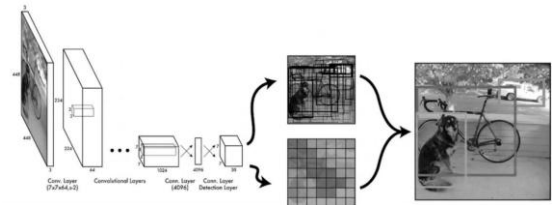


Fig. 1. Working of YOLO architecture

In an AI-based AOD pipeline, the tracker does not independently classify an object as abandoned; rather, it provides stable object and person tracks that are later evaluated by the AOD decision logic. For example, DeepSORT maintains track identities across consecutive frames by combining motion prediction with appearance-based data association [6].

The transition from an "owned" object to an "abandoned" object can then be described as a logical spatiotemporal rule. Firstly, a luggage item is associated with a nearby person when the distance between the person and the luggage remains within a predefined radius of association. Then, the system continuously evaluates whether the luggage remains stationary, whether the associated person moves outside this radius, and how long this condition persists. If the luggage remains below the movement threshold, the owner stays outside the association radius, and the stationary unattended state lasts longer than the predefined dwell-

time threshold, the object status changes from “owned” to “abandoned” [7].

Thus, the transition is recorded based on three tracked parameters: object/person track continuity, spatial separation between the luggage and its owner, and the duration of the stationary unattended state.

In AI methodology, particular attention is given to LE-DETR (Lightweight and Efficient Detection Transformer) models. LE-DETR is a lightweight end-to-end model for detecting abandoned objects on highways, consisting of three primary modules: RepELAN, TFusion, and CMI. These modules enhance the model's capability to detect abandoned objects in road environments, effectively accounting for significant differences in object scales while simultaneously reducing the high computational costs of the model [8].

Unlike CNNs, transformers utilize the self-attention mechanism, which allows for the effective modeling of relationships between objects at various distances. The Triple Fusion module within LE-DETR combines high-level semantic information with low-level spatial details, which is critical for detecting small abandoned items against the complex texture of a road or floor [8].

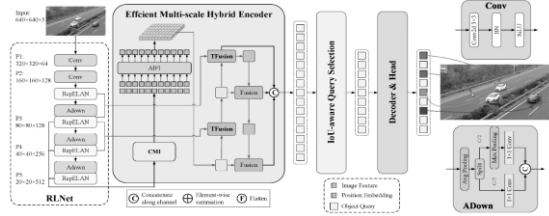


Fig. 2. Structure of the LE-DETR model

III. SCOPE OF WORK AND OBJECTIVES

The objective of this work is to conduct a comprehensive comparative analysis of abandoned object detection algorithms, which involves contrasting rule-based methods with artificial intelligence models. The study is aimed at evaluating the operational efficiency of these approaches in dynamic environments to determine the most effective solutions for modern intelligent video surveillance systems.

To achieve the stated objective, the following tasks have been defined:

- Analysis and classification of modern methods for automated object detection in video streams based on their operational principles and application conditions.
- Systematization of the fundamental differences between traditional background subtraction methods and modern neural network architectures.
- Identification of the advantages and limitations of each approach based on the criteria of accuracy, speed, and robustness against false positives.

- Formulation of recommendations for selecting optimal detection methods depending on available computational resources and the complexity of operational conditions.

The results of evaluating these parameters allow for the formation of a structured approach to selecting a detection strategy for various security scenarios. This work establishes the technical foundation for improving the reliability of automated monitoring in public spaces.

IV. EVALUATION OF RULE-BASED AND AI-ORIENTED APPROACHES

It is appropriate to divide the task of detecting abandoned luggage in video surveillance into two main phases. In the first phase, the identification of the luggage itself and its owner within the video frame is performed, while the second phase consists of evaluating the probability that the object is indeed abandoned based on the gathered information.

Traditional approaches to detecting luggage and its owner are largely based on background subtraction methods, which are simple to implement but ineffective for complex scenes with a large number of moving objects. One of the early methods involved the use of Markov Chain Monte Carlo (MCMC) models combined with Bayesian networks for luggage detection. Despite its effectiveness in controlled environments, this approach proved to be insufficiently adaptive to complex and dynamic settings.

Temporal background analysis has been applied to identify stationary objects. Although this approach is technically significant, it remains extremely sensitive to lighting variations and camera vibrations, leading to a high rate of false-positive detections. Another approach was based on foreground segmentation to identify immobile objects and generate candidates for abandoned luggage; while this reduced the number of erroneous detections, its robustness in real-world dynamic conditions remained insufficient.

More modern methods combine traditional algorithms with convolutional neural networks (CNNs), which are widely applied in computer vision tasks, particularly for classification, object detection, and image segmentation. Specifically, hybrid approaches have been proposed that combine background subtraction for static object detection with CNNs for the identification of abandoned luggage. Such solutions demonstrate high accuracy in controlled environments by reducing detection errors; however, their computational requirements remain a significant limitation for practical implementation.

Modern approaches are predominantly based on deep convolutional neural networks, which are well-suited for processing images and video from surveillance cameras. Object detectors are typically categorized into one-stage and two-stage models. In one-stage models, such as YOLO, object localization and classification are performed in a single pass through the network.

Conversely, two-stage approaches, notably Faster R-CNN, first generate region proposals and subsequently perform classification and coordinate regression, leading to higher computational costs and longer processing times. Due to their real-time capabilities and high accuracy in complex and dynamic scenes, models from the YOLO family are increasingly applied in security systems.

Furthermore, the emergence of Real-Time Detection Transformers (RT-DETR) has introduced a new paradigm that eliminates the need for non-maximum suppression (NMS), which is a common bottleneck in YOLO-based architectures. By utilizing an efficient hybrid encoder and uncertainty-aware query selection, RT-DETR achieves a superior balance between accuracy and latency, making it highly effective for identifying abandoned objects in high-density environments. Due to their real-time capabilities and high accuracy in complex and dynamic scenes, both the YOLO family and RT-DETR models are increasingly applied in security systems

A summary of relevant research is presented in Fig. 3 [7, 9].

Approach	Key Characteristics	Advantages	Disadvantages
Traditional BG Subtraction	Foreground segmentation (static/moving parts)	Low computational cost; ease of implementation	Low accuracy; sensitive to crowded scenes
MCMC and SVM	Bayesian networks + SVM classification	Effective tracking of short movements	Sensitive to environmental variations
Hybrid (CNN + BG)	Combination of background segmentation and CNNs	High accuracy; minimal false positives	Vulnerable to sudden light or scene changes
YOLO	One-stage detection + transfer learning	Maximum accuracy; adapts well to crowds	High GPU requirements for training
RT-DETR	End-to-end detector based on Transformers (ViT)	Real-time speed; no NMS needed; superior accuracy	Extremely high computational resource demands

Fig. 3. Comparison of object detection and tracking approaches

The comparison between YOLO and RT-DETR reflects the fundamental tension between traditional convolutional networks and modern transformers in computer vision tasks. For years, the YOLO series has remained the gold standard for real-time object detection due to its speed and efficient regression; however, it is still constrained by the necessity of post-processing results through Non-Maximum Suppression (NMS). The emergence of RT-DETR has been a game-changer, becoming the first transformer capable of competing with YOLO at high speeds by leveraging a global attention mechanism and eliminating redundant processing stages [9].

The figure presents a comparative analysis of the performance of three computer vision model architectures for the object detection task – YOLO26, YOLO11, and RT-DETRv2 – based on accuracy and speed metrics. The y-axis displays the COCO mAP (50–95) values, which characterize the generalized detection accuracy, while the x-axis represents the processing latency of a single image in milliseconds, measured on an NVIDIA T4 GPU using TensorRT 10. Lower latency

values correspond to higher model processing speeds [10].

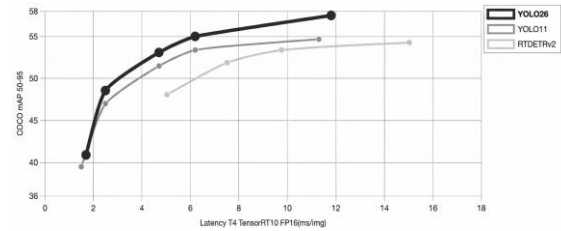


Fig. 4. Efficiency of YOLO Family and RT-DETR Models

The figure data are adapted from Ultralytics YOLO26 benchmark data and official performance metrics, including COCO mAP(50–95) and inference latency on an NVIDIA T4 GPU with TensorRT 10 [10].

Analysis of the graph indicates that the YOLO26 architecture demonstrates consistent dominance over other approaches across the entire studied speed range. At identical latency values, models in this series provide higher accuracy metrics compared to YOLO11 and RT-DETRv2. Specifically, the YOLO26x variant achieves a peak mAP value of approximately 57.5 with a latency of about 11.8 ms, signifying an optimal balance between recognition quality and computational costs.

Model	size (pixels)	mAP ^{val} 50-95	Speed CPU ONNX (ms)	Speed T4 TensorRT10 (ms)	params (M)	FLOPs (B)
RTDETRv2-s	640	48.1	-	5.03	20	60
RTDETRv2-m	640	51.9	-	7.51	36	100
RTDETRv2-l	640	53.4	-	9.76	42	136
RTDETRv2-x	640	54.3	-	15.03	76	259
YOLO26n	640	40.9	38.9	1.7	2.4	5.4
YOLO26s	640	48.6	87.2	2.5	9.5	20.7
YOLO26m	640	53.1	220.0	4.7	20.4	68.2
YOLO26l	640	55.0	286.2	6.2	24.8	86.4
YOLO26x	640	57.5	525.8	11.8	55.7	193.9

Fig. 5. Performance comparison: YOLO26 vs. RT-DETRv2

Figure 5 illustrates the performance differences between RT-DETRv2 and YOLO26. Note the significant disparity in CPU inference speed and parameter efficiency. The figure data are adapted from Ultralytics comparative benchmark data for YOLO26 and RT-DETRv2, including COCO mAP(50–95), TensorRT 10 latency, parameter count, and FLOPs [11].

Although the RT-DETRv2 architecture demonstrates significant scientific potential for transformer-based approaches in object detection tasks, the reviewed benchmark data indicate that the Ultralytics YOLO26 model family may be more appropriate for practical abandoned object detection scenarios. A comparative analysis of accuracy metrics (mAP 50–95), inference latency on the NVIDIA T4 GPU, and computational complexity (number of parameters and FLOPs) confirms the substantial advantage of YOLO26 in terms of the speed-to-quality ratio.

Low inference latency and stable performance on edge devices are particularly important for real-time abandoned object detection in video streams. In this context, YOLO26 models demonstrate significantly

Detection of Abandoned Objects in Video Surveillance Systems: a Comparative Analysis of Rule-Based and AI-Oriented Approaches

shorter frame processing times at comparable or higher accuracy levels than RT-DETRv2, which is critical for the timely identification of potentially hazardous situations in video surveillance systems.

The combination of an end-to-end NMS-free architecture, training optimization via MuSGD, and high inference efficiency makes YOLO26 a versatile and scalable solution for automated abandoned object detection systems – ranging from smart surveillance cameras to high-load analytical platforms. Thus, YOLO26 can be regarded as a promising and practice-oriented choice for implementation in real-world security systems that simultaneously require high accuracy, low latency, and industrial-grade readiness.

To summarize the reviewed evidence, this work presents two tables. Table 1 provides a comparative characterization of rule-based and AI-based methods across key performance indicators, including accuracy, occlusion robustness, lighting impact, computational complexity, and false-positive rates. Table 2 reflects the recommended application scenarios for the respective approaches depending on operational conditions, such as environment type, scene dynamics, and available computational resources [12].

Table 1

Evaluation of Rule-Based and AI-Based Approaches for Abandoned Object Detection in Video Surveillance

Evaluation Criterion	Rule-based methods (GMM, ViBE)	AI-based methods (YOLOv11, YOLOv26, RT-DETR)
Accuracy (mAP)	Low; depends on background clarity and absence of noise	High (0.75 - 0.99) due to semantic recognition
Occlusion Robustness	Minimal; occlusion resets the stationary state	High; trackers restore IDs after exiting occlusion
Lighting Impact	Critical; shadows and reflections create false objects	Moderate; mitigated by diversity in training data
Computational Complexity	Low; optimized for CPU and DSP	High; requires GPU/NPU for real-time inference
Frame Rate (FPS)	Very high (100+ FPS on standard hardware)	Medium (25-60 FPS depending on device power)
Setup Complexity	High; requires manual threshold tuning for each scene	Low for user; high for developer (data collection/labeling)
False Positives	High frequency due to dynamic background and artifacts	Low; ability to distinguish a person from a suitcase

The provided generalizations confirm that rule-based methods are appropriate for stable and resource-constrained environments, whereas AI-based solutions, particularly YOLO26, are more effective for complex, dynamic scenes and real-time systems where high

accuracy, low inference latency, and scalability are critical.

Table 2

Scenario-Based Selection of Rule-Based vs. AI-Based Detection Methods

Application Scenario	Recommended Approach	Description / Rationale for Selection
Indoor spaces with controlled lighting (warehouses, technical corridors, server rooms)	Rule-based	Minimal scene dynamics. A simple algorithm (e.g., GMM + temporal threshold) ensures high processing speed (FPS) and reliability with minimal resources.
Systems with strict budget constraints (legacy facilities, large number of cameras)	Rule-based	Classical methods can operate on existing video recorders (DVR/NVR) without the need to replace infrastructure with expensive server capacities.
High-density passenger traffic areas (airports, railway stations, subways)	AI (Deep Learning)	Requires tracking the owner through crowds and understanding the semantics of objects and people. AI detectors with trackers (DeepSORT, ByteTrack) handle this complexity better.
Open spaces and complex weather conditions (plazas, parking lots)	AI (Deep Learning)	Changes in lighting, rain, or snow render classical background subtraction inoperable. Neural networks provide significantly higher accuracy.
Requirements for detailed investigation	AI (Deep Learning)	Allows not only for object detection but also for identifying its type, color, and other attributes for rapid transmission to response teams.

V. CONCLUSION

Based on the comprehensive analysis of rule-based and AI-oriented approaches for abandoned object detection in video surveillance systems, it can be concluded that the selection of an optimal method is determined by scene conditions, accuracy requirements, and available computational resources. As no universal solution exists, a scenario-oriented or combined approach is the most effective.

Rule-based methods, specifically background subtraction based on GMM or ViBE, are appropriate for simple and stable environments: indoor spaces with controlled lighting, minimal dynamics, and low pedestrian density. Their primary advantages include exceptionally high processing speeds and the ability to operate without a GPU, making them suitable for budget-constrained systems or legacy infrastructure. However, these approaches remain vulnerable to lighting

changes, occlusions, and noise, which leads to a high rate of false positives in real-world dynamic scenes.

AI-based approaches, built on deep neural networks, are more effective for complex environments with high crowd density, open spaces, and variable weather conditions. They provide high detection accuracy, semantic differentiation between persons and luggage, and robustness against occlusions through the use of tracking algorithms. Such methods allow not only for the detection of an abandoned object but also for the extraction of additional attributes necessary for response protocols. A comparison of modern detectors showed that in practical real-time scenarios, models from the YOLO family, specifically YOLO26, demonstrated a superior balance between accuracy and inference latency compared to transformer-based approaches like RT-DETR. This made them more suitable for implementation in industrial video analytics systems and edge devices.

The most practical solution is a hybrid approach, where rule-based methods are utilized for the rapid detection of static candidates, while AI detectors perform semantic verification and spatiotemporal behavior analysis. This combination reduces computational load and the number of false positives while maintaining high accuracy. Ultimately, for real-world abandoned object detection systems, AI-based or hybrid solutions utilizing YOLO26 represent the most justified choice.

VI. CONFLICTS OF INTEREST

The author declares that there are no conflicts of interest regarding the publication of this paper.

VII. DECLARATION ON GENERATIVE AI

During the preparation of this work, the author used Grammarly for grammar and spelling correction, as well as for paraphrasing and rewording. All generated suggestions were carefully reviewed and, where necessary, revised by the author, who assumes full responsibility for the final content of the publication.

References

- [1] Lin, K., Chen, C., Hsieh, J., & Fan, K. (2015). Abandoned object detection via temporal consistency modeling and back-tracing verification for visual surveillance. *IEEE Transactions on Information Forensics and Security*, 10(7), 1359–1370. <https://doi.org/10.1109/TIFS.2015.2408263>
- [2] Viroli, C., & McLachlan, G. J. (2019). Deep Gaussian mixture models. *Statistics and Computing*, 29(1), 43–51. <https://doi.org/10.1007/s11222-017-9793-z>
- [3] Liu, H., Wei, H., Yang, G., Xia, C., & Zhao, S. (2023). An improved vibe algorithm based on adaptive thresholding and the deep learning-driven frame difference method. *Electronics*, 12(16), 3481. <https://doi.org/10.3390/electronics12163481>
- [4] Tsirtsakis, P., Zacharis, G., Maraslidis, G. S., & Fragulis, G. F. (2025). Deep learning for object recognition: A comprehensive review of models and algorithms. *International Journal of Cognitive Computing in Engineering*. <https://doi.org/10.1016/j.ijcce.2025.01.004>
- [5] Yilmazer, M., & Karakose, M. (2025). A new, robust, adaptive, versatile, and scalable abandoned object detection approach based on DeepSORT, dynamic prompts, and customized LLM for smart video surveillance. *Applied Sciences*, 15(5), 2774. <https://doi.org/10.3390/app15052774>
- [6] Wojke, N., Bewley, A., & Paulus, D. (2017). Simple online and realtime tracking with a deep association metric. 2017 IEEE International Conference on Image Processing (ICIP), 3645–3649. <https://doi.org/10.1109/ICIP.2017.8296962>
- [7] Vrsalovic, I., Lerga, J., & Ivasic-Kos, M. (2025). A system for real-time detection of abandoned luggage. *Sensors*, 25(9), 2872. <https://doi.org/10.3390/s25092872>
- [8] Zhang, B., Wang, C., & Wang, J. (2025). A lightweight and efficient detection transformer for highway abandoned objects. *Algorithms*, 18(3), 133. <https://doi.org/10.3390/a18030133>
- [9] Zhou, L., & Xu, J. (2024). Enhanced abandoned object detection through adaptive dual-background modeling and SAO-YOLO integration. *Sensors*, 24(20), 6572. <https://doi.org/10.3390/s24206572>
- [10] Hafeez, H., Garratt, M., Pletsted, J., Iyer, S., & Sowmya, A. (2025). YOLO11-4K: An Efficient Architecture for Real-Time Small Object Detection in 4K Panoramic Images. arXiv preprint arXiv:2512.16493. <https://doi.org/10.48550/arXiv.2512.16493>
- [11] Sapkota, R., & Karkee, M. (2025). Ultralytics YOLO evolution: An overview of YOLO26, YOLO11, YOLOv8 and YOLOv5 object detectors for computer vision and pattern recognition. arXiv preprint arXiv:2510.09653. <https://doi.org/10.48550/arXiv.2509.25164>
- [12] Ali, M. L., & Zhang, Z. (2024). The YOLO framework: A comprehensive review of evolution, applications, and benchmarks in object detection. *Computers*, 13(12), 336. <https://doi.org/10.3390/computers13120336>



Vitalii-Oleksandr Pashukh was born in Lviv region, Ukraine. He began his professional path at Sheptytskyi Mining and Economic College, specializing in Software Development. Later, he earned a Bachelor's degree in Software Engineering at Lviv Polytechnic National University. Currently, he is advancing his expertise as a Master's student at IT STEP University. With a strong background in engineering, he specializes in the Insurance domain, focusing on building robust solutions and scalable systems.



Anastasiia Deineko, Ph. D. & Associate Professor, AI Solution Architect & Advisory Partner with 10+ years of experience in designing and delivering AI-driven solutions across FinTech, HealthTech, E-commerce, Manufacturing, and Security. Currently, she is focused on solution architecture for large-scale AI systems, bridging advanced research with business-ready applications. Proven ability to drive both technical implementation and strategic adoption of AI technologies.